ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q2, Q3, Q4

Question 2

(a) States $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau - b\,x_2 - mg\ell \sin x_1]$.

Equilibria ($\tau = \tau_0$, $\dot{x} = 0$): $x_2 = 0$ and $mg\ell \sin\theta_e = \tau_0$. Sence $|\sin\theta| \le 1$, an equilibrium exists iff $|\tau_0| \le mg\ell = 2.4525$ N·m.

(b) Jacobian at $\theta_e = 60°$: $A = [0,\ 1;\ -(mg\ell/J)\cos\theta_e,\ -b/J]$.

$mg\ell/J = 2.4525/0.02 = 122.625$; $\cos(60) = -0.9524$, so $A_{21} = -122.625 \times (-0.9524) = +116.8$. $A_{22} = -b/J = -2.5$.

$A = [0,\ 1;\ 116.8,\ -2.5]$, **$B = [0;\ 50]$**.

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$, so $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$, i.e. $50(90 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$.

Gravity torque is non-zero at 90°, so the left side can't be zero, hence $\theta_{ss} \neq \theta_d$.

Taking $\sin\theta_{ss} \approx 1$: $90 - \theta_{ss} = 2.4525/50 = 0.049$, so the stedy-state error ≈ **0.049°**.

(d) With compensation, at rest: $K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss} = mg\ell \sin\theta_{ss} \Rightarrow \theta_{ss} = \theta_d$: **steady-state error = 0**.

Closed loop: $J\ddot{\theta} + (b + K_d)\dot{\theta} + K_p(\theta - \theta_d) = 0$; gravity cancels. Without compensation the linearised stiffness is $K_p + mg\ell\cos\theta_d$, and at $\theta_d = 90°$ $\cos\theta_d = 0$, so the stiffness is K_p either way. **Poles unchanged** (specific to 90°; at other angles they would shift).

Question 3

(a) (1) Differentiation is a high-pass operation (gain ∝ ω), so encoder measurement/quantisaton noise is amplified, by about 1/T = 1000 at 1 kHz.

(2) Encoder quantisaton: Δθ is a whole number of counts, so $\hat{\omega} = \Delta\theta/T$ moves in steps of (count size)/T; at low speed the estimate is a coarse staircase of zeros and spikes.

(b) $H(s) = \omega_c/(s + \omega_c)$, $\omega_c = 2\pi(20) = 125.66$ rad/s. Tustin: $s = (2/T)(z - 1)/(z + 1)$, T = 1 ms. With $a = \omega_c T/2 = 0.06283$: $H(z) = a(1 + z^{-1}) / [(1 + a) - (1 - a)z^{-1}]$.

$y_k = [a/(1+a)](u_k + u_{k-1}) + [(1-a)/(1+a)]\, y_{k-1}$ = **$0.05912\,(u_k + u_{k-1}) + 0.88177\,y_{k-1}$**.

DC gain: z = 1 gives $H(1) = 2(0.05912)/(1 - 0.88177) = 0.11824/0.11823 = 1.000$ ✓.

(c) $A - LC = [-\ell_1,\ 1;\ -\ell_2,\ -2]$. $\det(sI - A + LC) = (s + \ell_1)(s + 2) - \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 - \ell_2)$.

Desiered $(s + 10)^2 = s^2 + 20s + 100$. So $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$ and $2\ell_1 - \ell_2 = 100 \Rightarrow \ell_2 = 36 - 100 = -64$.

$L = [18,\ -64]^T$.

(d) Observer advantage: becuase it is a copy of the plant corrected by the encoder error, it gives a velocity estimate without the delay that a filter needs for noice rejection, and it can also estimate quantities no filter on θ can, such as an unknown load torque.

A filter is the safer choice when the model is wrong or the true input is not known, e.g. unmodelled friction or backlash, a changing payload, or acuator saturation: the obsever would then be driven by a wrong u and drift, while the filter only uses measured data.

Question 4

(a) Closed loop: $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\, r$, i.e. $s^2 + 4s + 8 = 0$.

$s = [-4 \pm \sqrt{(16 - 32)}]/2 = -4/2 \pm \sqrt{(-16)} = \mathbf{-2 \pm 4j}$.

DC gain $= N_{ff}/8 = 1$, so $\mathbf{N_{ff} = 8}$.

(b) With $b = 0.75$ ($N_{ff} = 8$, $k_1 = 2$): $G_{DC} = bN_{ff}/(c + bk_1) = 0.75 \times 8/(6 + 0.75 \times 2) = 6/7.5 = 0.8$.

$\theta_{ss} = \mathbf{0.8}$. Tracking error $= (1 - 0.8)/1 \times 100 = \mathbf{20\%}$. (Less than the 25% acuator loss, because the stiffness also dropped.)

(c) Substituting u into the plant and using $\dot{\xi} = -\theta$ ($r = 0$): characteristic polynomial $s^3 + (a + k_2)s^2 + (c + k_1)s + K_I$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

$2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$; $6 + k_1 = 24 \Rightarrow \mathbf{k_1 = 18}$; $\mathbf{K_I = 32}$.

(d) In stedy state $\dot{\xi} = r - \theta_{ss} = 0$, so $\theta_{ss} = r$ whatever b and c are, provided the loop is stable.

Failure mode: integrator windup — when the acuator saturates the integral keeps growing and causes a big overshoot later.